

Week 1 (Tuesday, June 17)

Course Welcoming and Introduction

CMPE-257: Machine Learning - Summer 2025



SJSU SAN JOSÉ STATE
UNIVERSITY

Instructor: Rafael E Banchs
Email: rafael.banchs@sjsu.edu

Today's plan...

Main Topic: Introduction to Machine Learning

5:30-6:00 Introductions

6:00-6:40 Session 1

6:40-7:00 Break

7:00-9:00 Session 2

9:00-9:30 Wrap-up and HW1

Session 1:

- Syllabus and course work plan

Session 2:

- Introduction to Machine Learning
- Hands-on coding session 1

Introductions...

- Your name (nickname?)
- Your company/organization
- What do you do?
- Why are you here?
- What do you expect?



Week 1 - Session 1

Syllabus and Course Work Plan

- Goals and learning outcomes
- Course evaluation plan
- Grading policy
- Timeline and work plan

Course Goals

- CG1: To demonstrate an understanding of advanced knowledge of the practice of computer/software engineering
- CG2: To tackle complex engineering problems and tasks, using contemporary engineering principles, methodologies and tools
- CG3: To demonstrate leadership and the ability to participate in teamwork in an environment with different disciplines of engineering
- CG4: To advance successfully in the engineering profession, and sustain a process of lifelong learning
- CG5: To be aware of ethical, economic and environmental implications of ML

Course Learning Outcomes (CLOs)

- CLO1: Understand and explain fundamental concepts, practices, tools and methods of Machine Learning
- CLO2: Become familiar with training algorithms and evaluation methods applied to different families of linear and non-linear ML models
- CLO3: Apply modern tools and programming languages to perform ML tasks in fields such as science, engineering, healthcare, social media, etc. with the appropriate techniques
- CLO4: Design, implement, and present a team project focusing on ML applications, gaining knowledge from their own and other classmate project work and approaches

Evaluation Plan: High-level Overview

Activity	Comments	Total Weight
Homeworks	10 individual assignments	20%
Term Project	Teams of 3 or 4 students	15%
Quizzes	3 quizzes on Canvas	15%
Mid-term exam	120 mins on July 24	25%
Final exam	120 mins on September 04	25%

Evaluation Plan: Homeworks (20%)

- There will be ten individual take-home assignments (2% each)
- Assignments mostly include programming sections but some might include written sections
- Assignments are due on 11:59pm PST of the corresponding due date
- No late submissions are allowed. Please submit whatever you have done by the due date and time!

Evaluation Plan: Term Project (15%)

- Teams of three or four students will be formed to work on a term project
- The project topic must be related to the topics discussed in class
- Terms projects will be graded based on: project proposal (5%), final project presentation (5%), and final report (5%)
- More detailed requirements on project phases will be provided later

Evaluation Plan: Quizzes (15%)

- There will be three in-class quizzes in total (5% each)
- Each quiz will contain:
 - 10 multiple selection questions (2 points each)
 - 5 short problems (6 points each)
- All quizzes are open book and conducted on Canvas
- **No make-ups are allowed !**

Evaluation Plan: Mid-term Exam (25%)

- A 120-min mid-term exam will be conducted on July 24
- The mid-term exam will account for 25% of the final grade
- The exam will include multiple choice and short problem questions
- The exam evaluates materials covered until July 17, inclusive.
- The exam is open book and will be conducted on Canvas
- **No make-up of the mid-term exam is allowed !** (If a student has serious/compelling reasons that can be proven by an independent authority, the final exam will be graded at 50% for that student).

Evaluation Plan: Final Exam (25%)

- A 120-min final exam will be conducted on September 04
- The final exam will account for 25% of the final grade
- The exam will include multiple choice and short problem questions
- The exam evaluates all materials covered in class after the mid-term
- The exam is open book and will be conducted on Canvas
- **No make-up of the final exam is allowed !** (If a student has serious/compelling reasons for missing the exam that can be proven by an independent authority, and exception can be made and a new date will be arranged for taking the final exam)

Grading Policy

A+	$\geq 98\%$
A	$\geq 94\%$
A-	$\geq 90\%$

B+	$\geq 88\%$
B	$\geq 84\%$
B-	$\geq 80\%$

C+	$\geq 78\%$
C	$\geq 74\%$
C-	$\geq 70\%$

D	$\geq 60\%$
F	$< 60\%$

Course Timeline: First Half

Week	Date	Contents	Evaluation
1	6/17	Introduction to Machine Learning	-
2	6/26	Statistical Learning Theory	HW1 due
-	7/03	National Holiday Week	-
3	7/10	Linear Regression and Classification	HW2 due, Quiz 1
4	7/17	Overfitting, Regularization, Validation	HW3 due
5	7/24	Term Project Proposals	HW4 due, Mid-term

Course Timeline: Second Half

Week	Date	Contents	Evaluation
6	7/31	Polynomial Regressions	HW5 due
7	8/07	Tree-based Methods	HW6 due, Quiz 2
8	8/14	Support Vector Machines	HW7 due
9	8/21	Neural Networks and Deep Learning	HW8 due, Quiz 3
10	8/28	Term Project Presentations	HW9 due
11	9/04	Advanced Topic (Invited Lecture)	HW10 due, Final

Course sessions and work dynamics

- An interactive program, including exercises and coding sessions
- Proactive participation is expected and highly encouraged
- Quizzes and exams will be conducted at the beginning of the session
- Wrap-up and targeted Q&A sessions will be conducted at the end
- Sessions will typically include one long break (~20 mins) and a few short breaks (3 to 5 mins each)
- You are expected to be in front of a computer during the full session!

Course materials and resources

- There is a list of references and reading materials available
- Each session will provide additional pointers to specific materials
- Notebooks with python code and exercises are also provided
- Slide decks are also available (pay attention to the special markers)

Take Note!



A fundamental concept is being introduced

Review!



Relevant concepts are revisited and clarified

Exercise!



An opportunity for practice is provided

More about computational resources

- You will need Python 3, as well as some specific python modules, installed in your personal computer or laptop
- In-class coding sessions and home assignments must be conducted on Jupyter Notebooks and are designed to fit in a single computer
- You might need large scale computational resources for your Project, in such case, you can get access to the CoE-HPC infrastructure:
 - I need to submit the request in your behalf. So, let me know!
 - For more info, see: <https://www.sjsu.edu/cmpe/resources/hpc.php>

Contact information and office hours

- Email: rafael.banchs@sjsu.edu
- Text and/or Whatsapp: 650-665-1046
- Session hours: Thursdays from 5:30pm to 9:30pm
- Office hours: Tuesdays from 5:30pm to 6:30pm
- Instructional Student Assistant: Dhruv Khut



Week 1 - Session 2

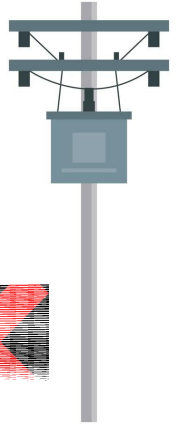
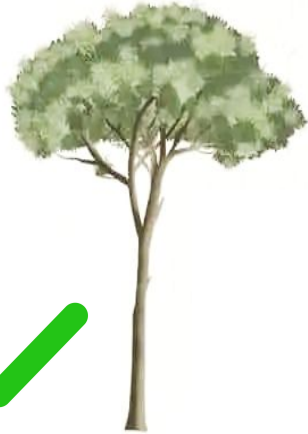
Introduction to Machine Learning

- Learning from data
- The three learning paradigms
- A supervised learning example
- Machine learning in practice

How do we learn?



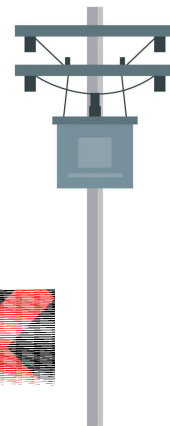
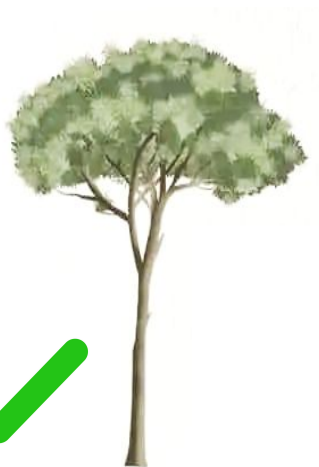
Which of these objects is a tree?



How do we learn?



Can you explain your answer?



How do we learn?

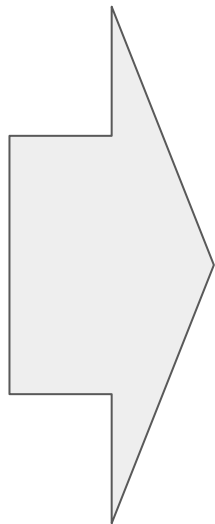
- There are different learning mechanisms
- For the one discussed here... we learn from data!
 - Exposure to examples of **data** points is required
 - We are able to identify **patterns** in the data
 - The patterns are used to create a mental **representation**
 - The mental representation is used to **analyze** new data

Learning from data...



Train Data

Learning



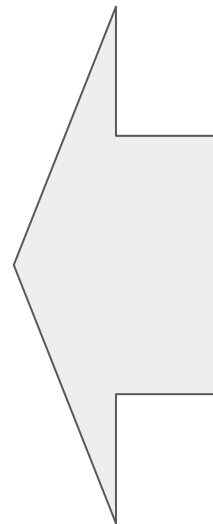
Model



Test Data



Inference



Learning from data...

- Train data: a set of data samples used for learning
- Learning: using an algorithm to learn a model
- Model: a usable representation of the data space
- Inference: using a trained model to analyze new data
- Test data: any new data not used for learning
- Evaluation: using new data to assess model performance

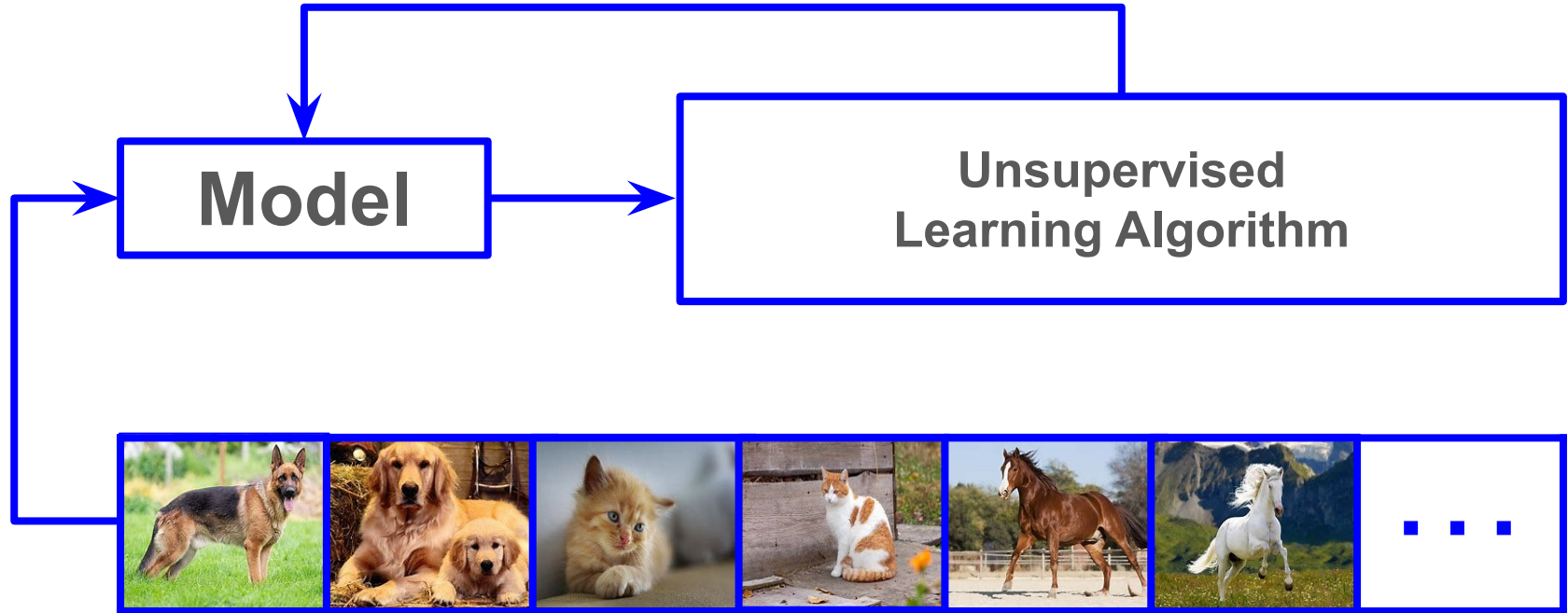


The three learning paradigms

There are three fundamental learning paradigms in machine learning:

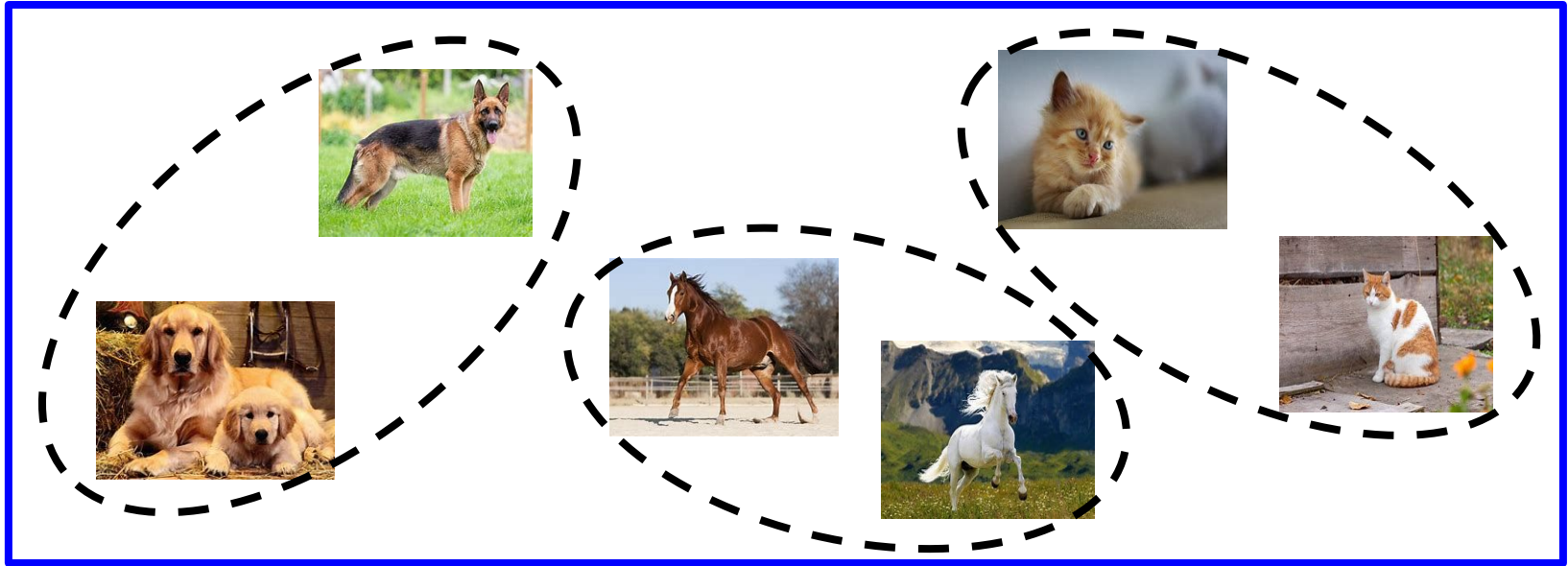
- **Unsupervised Learning (UL)**
 - Aims at identifying latent structures and patterns within the data
- **Supervised Learning (SL)**
 - Aims at reducing the error between actual and predicted values
- **Reinforcement Learning (RL)**
 - Aims at finding an optimal strategy to complete a multi-step task

Unsupervised learning in a nutshell

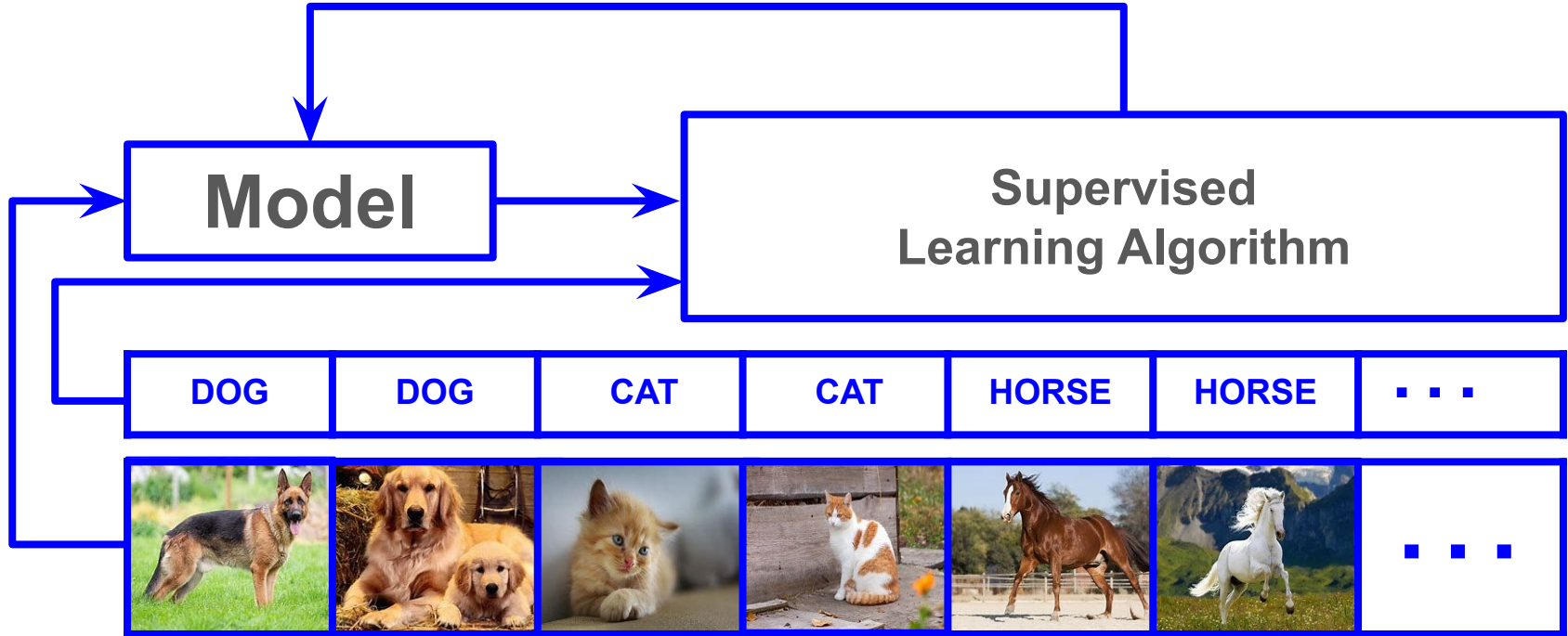


Unsupervised learning in a nutshell

Feature Space (model's internal data representation)

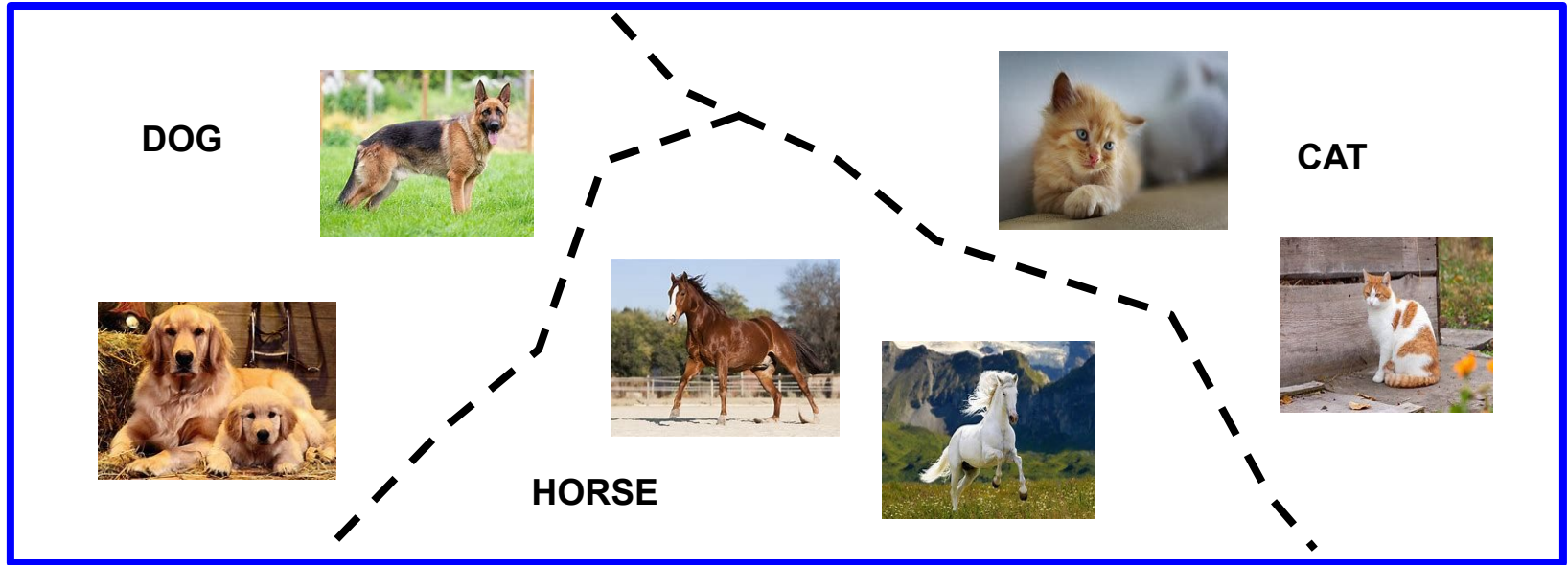


Supervised learning in a nutshell

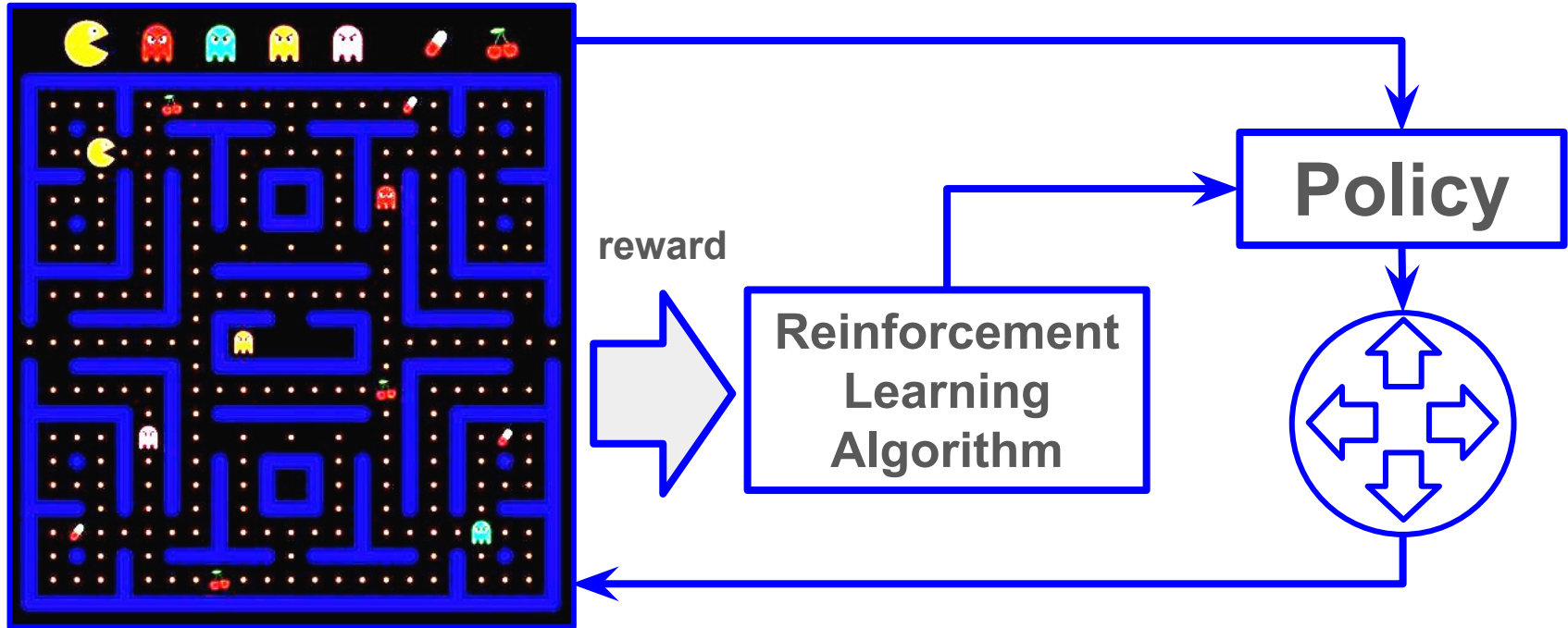


Supervised learning in a nutshell

Feature Space (model's internal data representation)

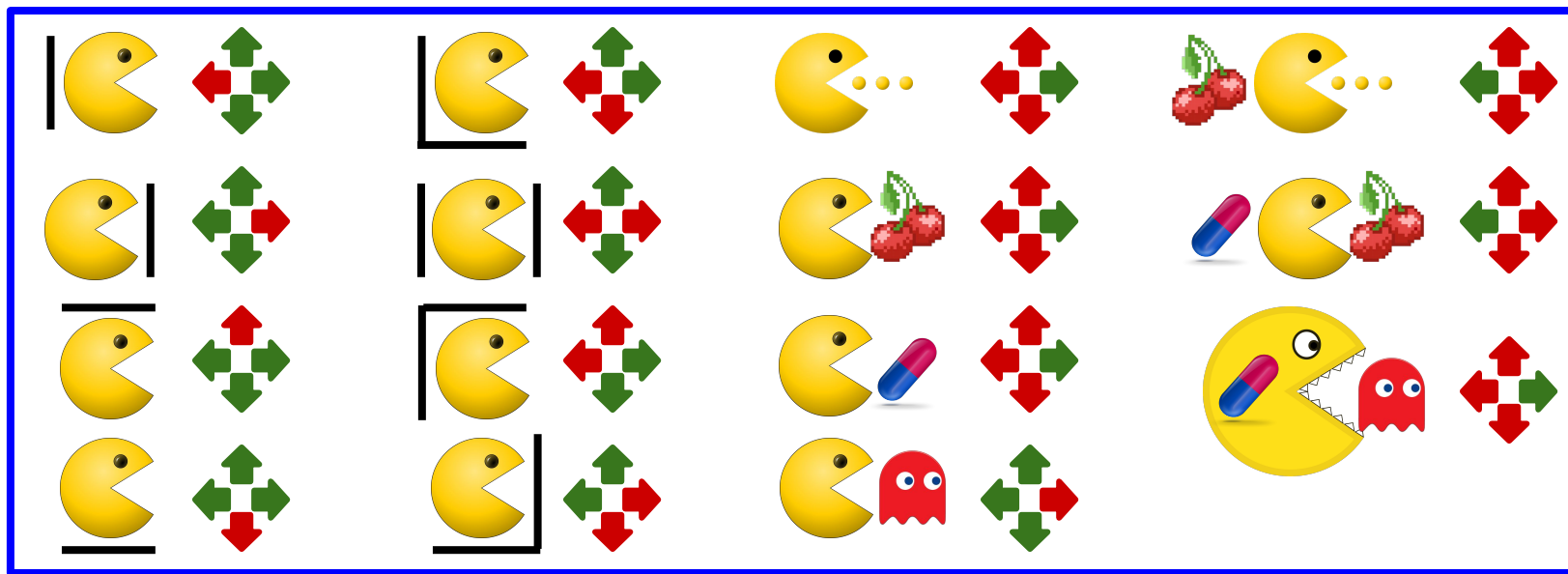


Reinforcement learning in a nutshell



Reinforcement learning in a nutshell

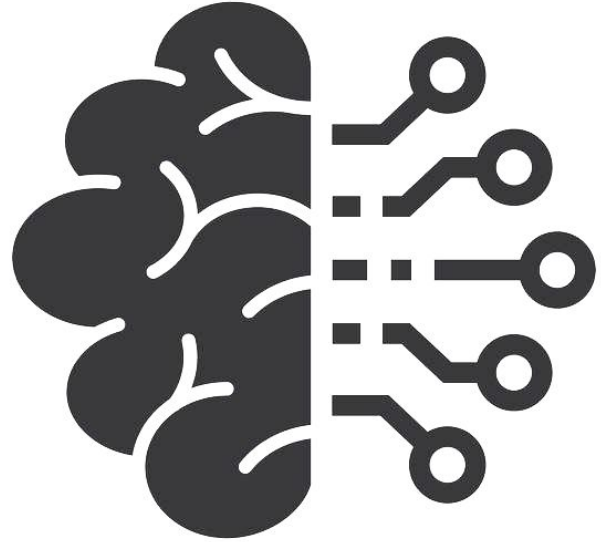
Action Space (model's internal policy)



Disclaimer...

This course is mainly on Supervised Learning!

- Linear regression
- Supervised classification
- Polynomial regression
- Three-based methods
- Support vector machines
- Neural networks and deep learning



Supervised versus unsupervised learning

- Unsupervised learning:
 - Aims at identifying latent structures within the data
 - Do not require the availability of target values or categories
 - Ideal for exploratory analysis and relationship discovery
- Supervised learning:
 - Aims at reducing prediction errors
 - Require the availability of target values or categories
 - Ideal for predictive tasks requiring estimation of outcomes



Types of machine learning models

- Based on the type of outputs they generate
 - Classification vs Regression
- Based on the assumptions made on $f(x)$ properties
 - Parametric vs Non-parametric
- Based on the type of input/output relation assumed
 - Linear vs Non-linear
- Based on the type of probabilities they attempt to learn
 - Discriminative vs Generative

Regression versus classification models

- Regression models:
 - Are used to predict continuous target variables
- Classification models:
 - Are used to predict discrete/categorical target variables
 - Multi-class classification:
 - Target categories/labels are mutually exclusive
 - Multi-label classification:
 - Target categories/labels are not mutually exclusive

Parametric versus non-parametric models

- Parametric models:
 - Assume a predefined underlying data distribution
 - Are computationally efficient and highly interpretable
 - Have limited flexibility to capture different data distributions
- Non-parametric models:
 - Do not make assumptions about data distributions
 - Exhibit flexibility to capture complex data distributions
 - Are computationally complex and difficult to interpret

Linear versus non-linear models

- Linear models:
 - Assume a linear association between inputs and outputs
 - Are mathematically simple and highly interpretable
 - Cannot capture the complexity in most real world problems
- Non-linear models:
 - Do not assume a linearity between inputs and outputs
 - Offer more flexibility to capture complex relationships
 - Are computationally complex and exhibit low interpretability

Discriminative versus generative models

- Discriminative models:
 - Focus on learning decision boundaries between classes
 - Base their predictions on conditional probabilities $p(y|x)$
 - Commonly used for supervised learning tasks
- Generative models:
 - Focus on learning how to produce data samples
 - Base their predictions on joint probabilities $p(y,x)$
 - Commonly used for unsupervised learning tasks

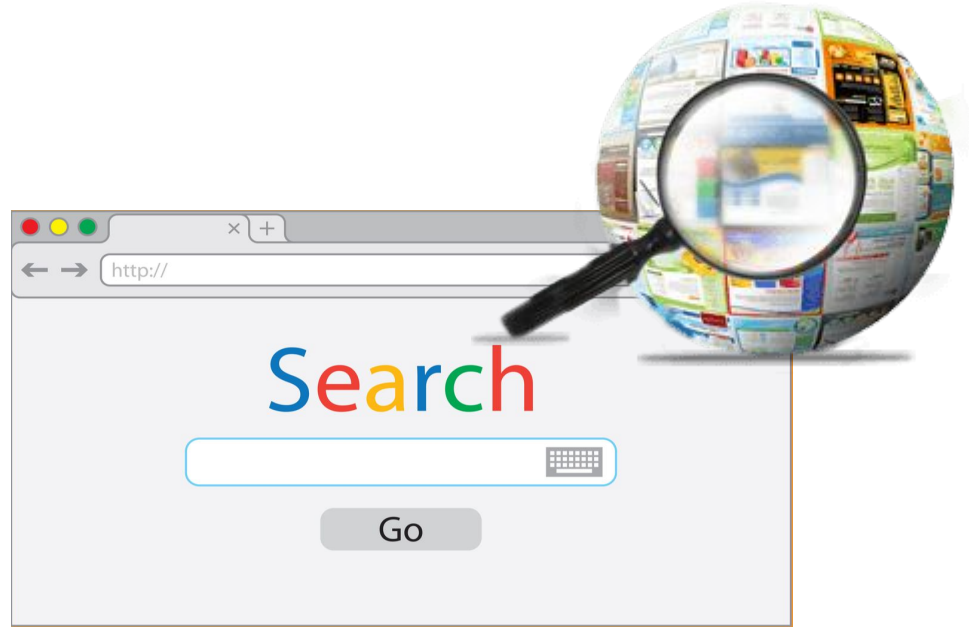
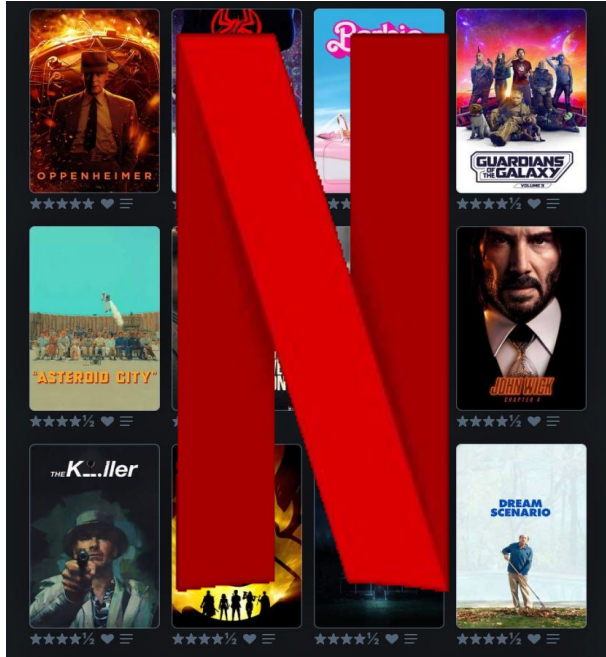


Types of machine learning problems

- Association: assessing similarities and differences
- Classification: assigning labels/tags to data points
- Regression: estimating a numerical valued output
- Generation: producing a new data sample
- Decision: making decisions based on contextual information

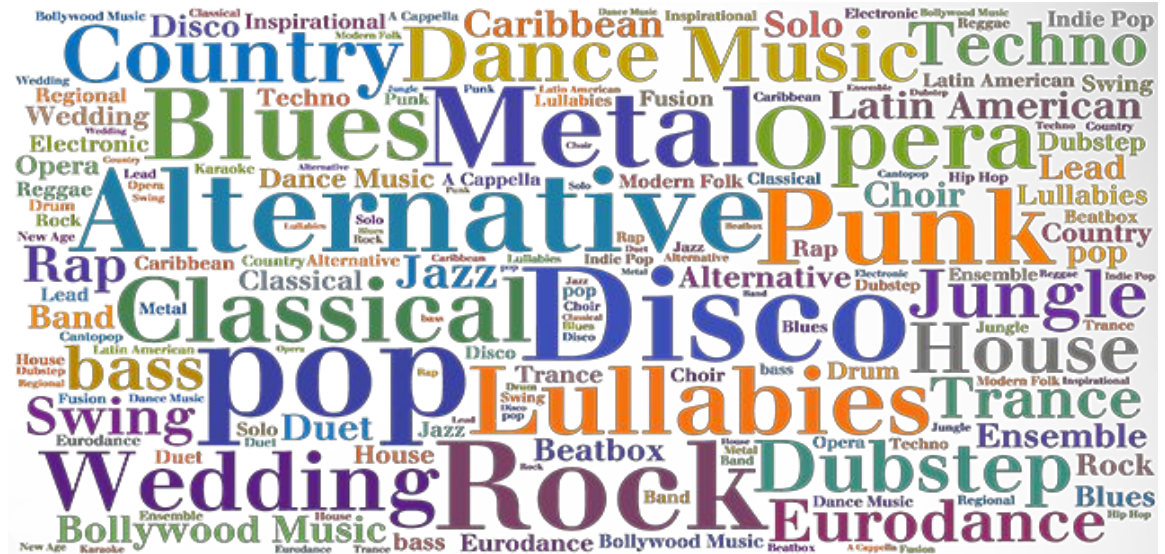
Examples of association problems

- Content and product recommendations, web search, etc.



Examples of classification problems

- News and email classification, content labelling, etc.



Examples of regression problems

- Price estimation, risk probability estimation, etc.



Examples of generation problems

- Machine translation, speech synthesis, image generation, etc.

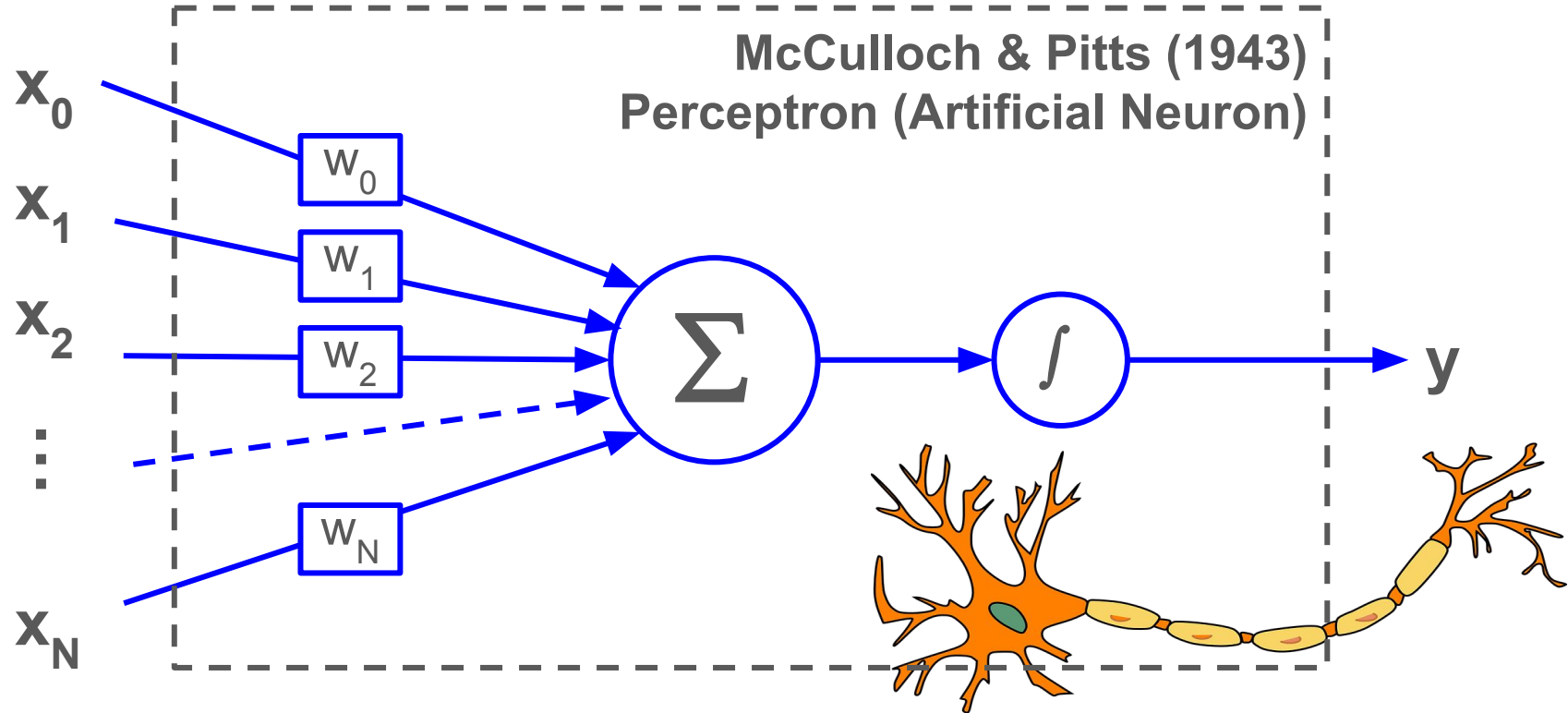


Examples of decision problems

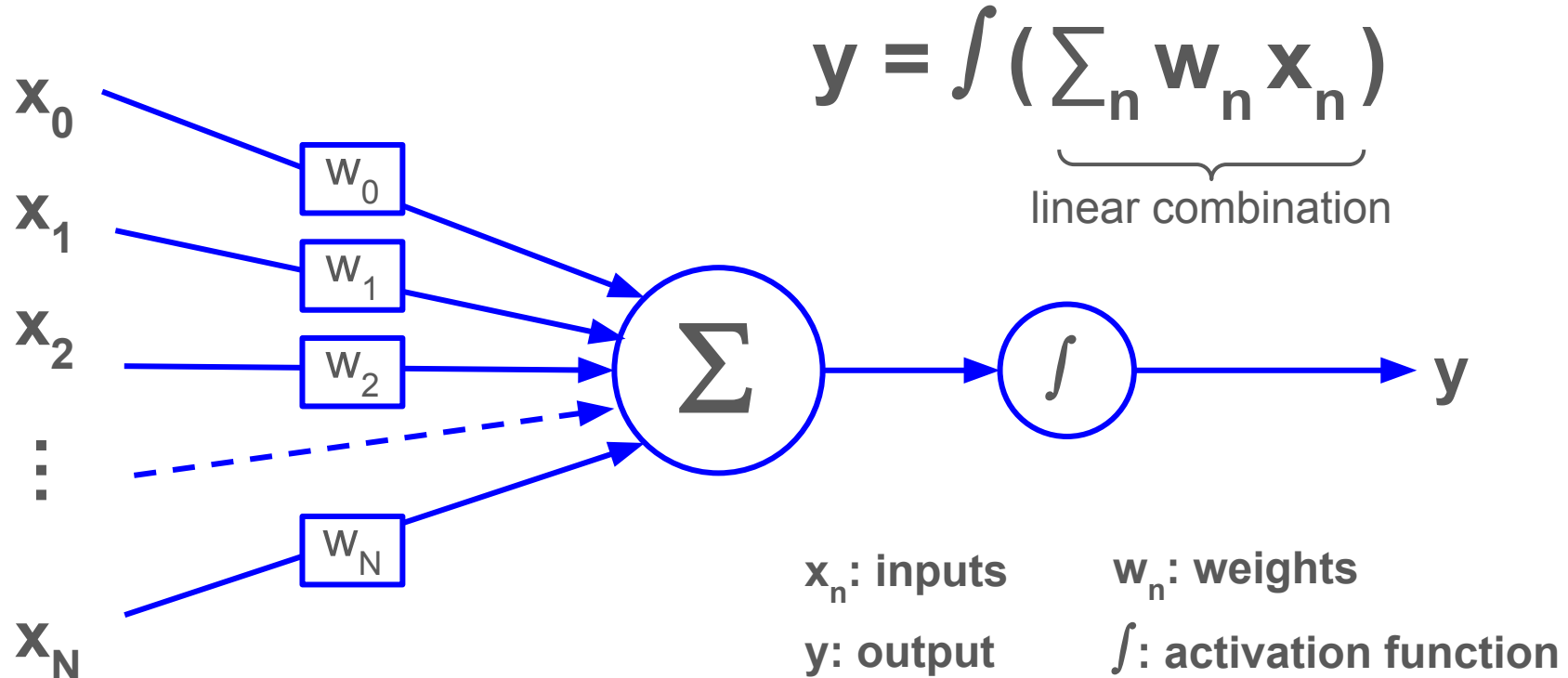
- Dialog systems, autonomous driving, robot navigation, etc.



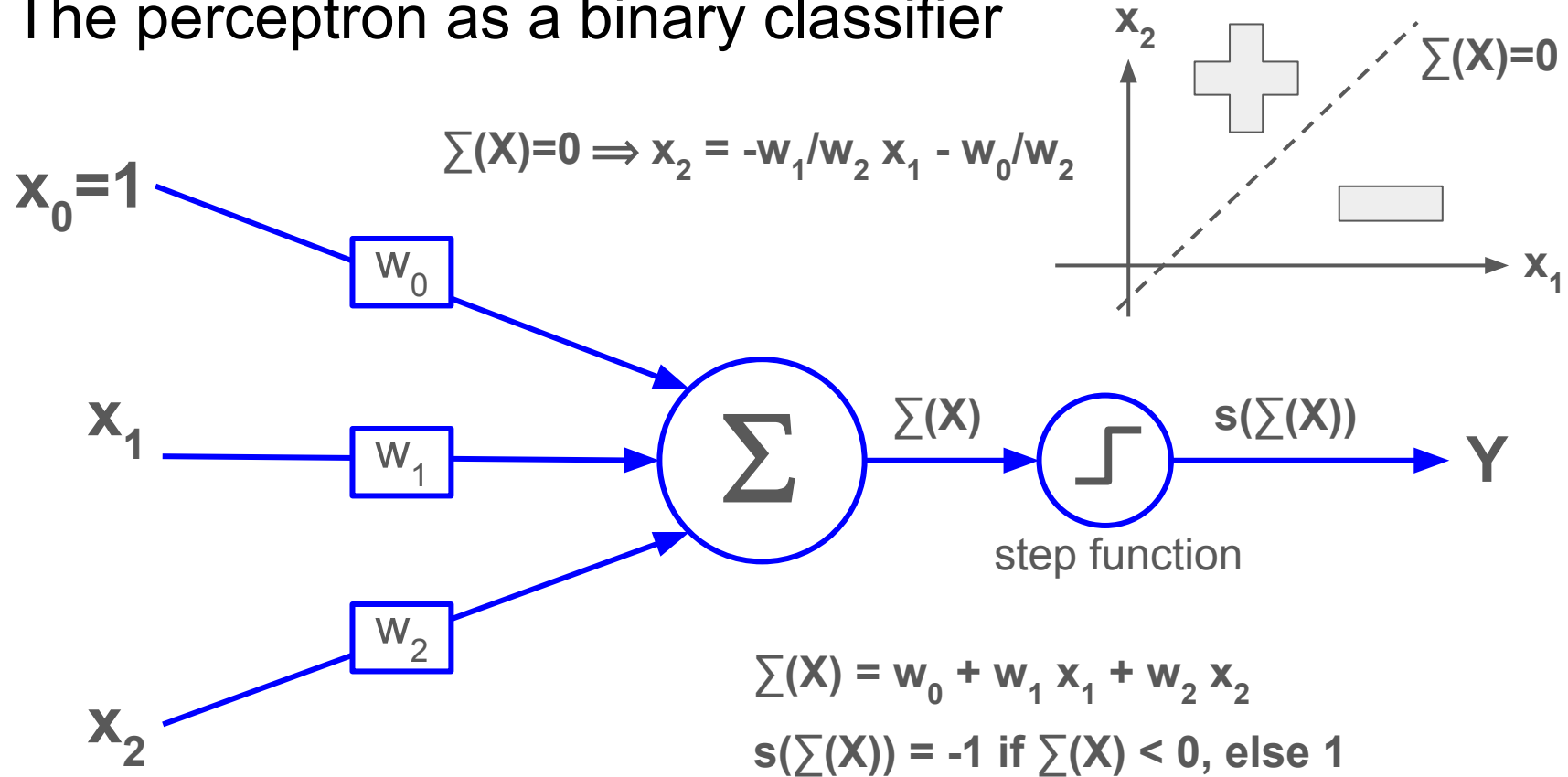
The perceptron: a model for an artificial neuron



The perceptron: a linear combinator with an activation

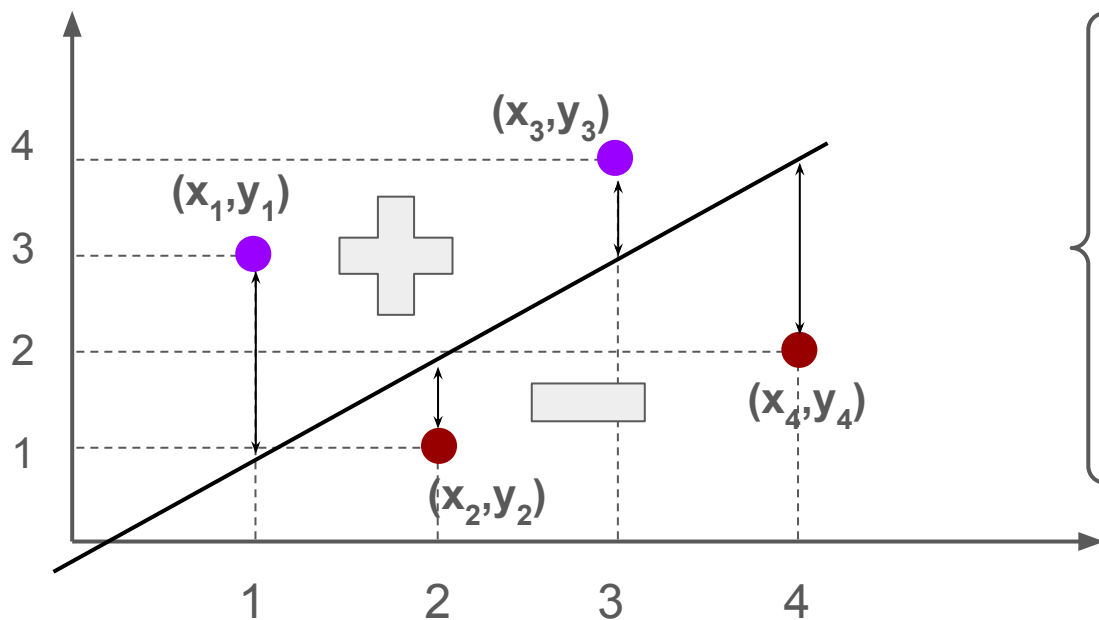


The perceptron as a binary classifier



Analytical solution for a linear classifier in $\mathbb{R}^2$

We want to find a line $y = m x + b$ that separates two data classes

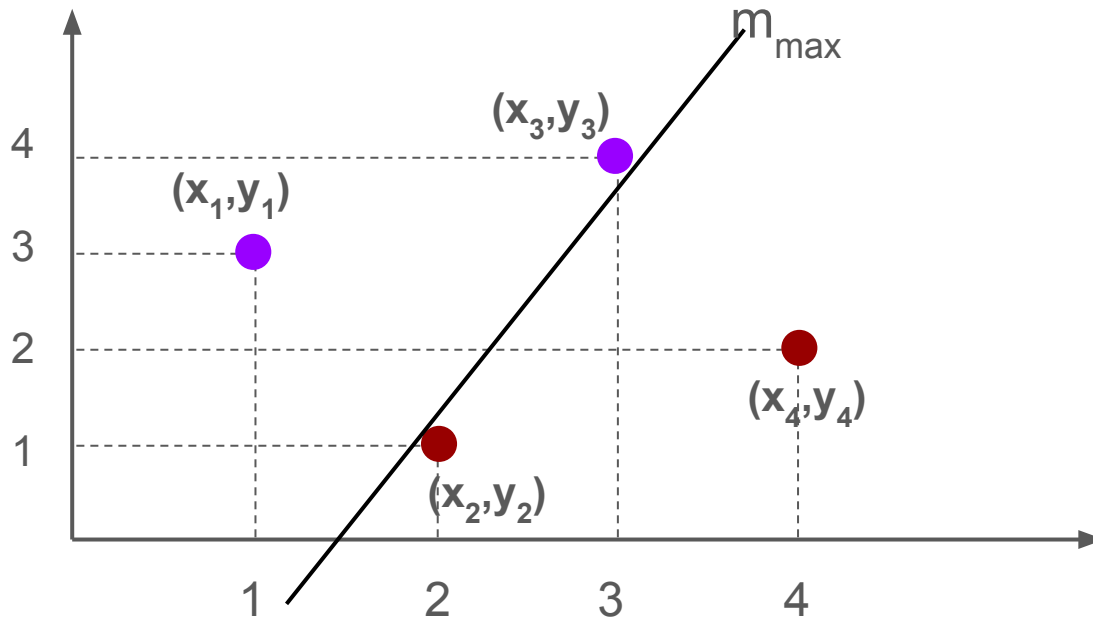


$$\begin{cases} y_1 - (m x_1 + b) > 0 \\ y_2 - (m x_2 + b) < 0 \\ y_3 - (m x_3 + b) > 0 \\ y_4 - (m x_4 + b) < 0 \end{cases}$$

$m = ? \quad b = ?$

Analytical solution for a linear classifier in $\mathbb{R}^2$

We want to find a line $y = m x + b$ that separates two data classes



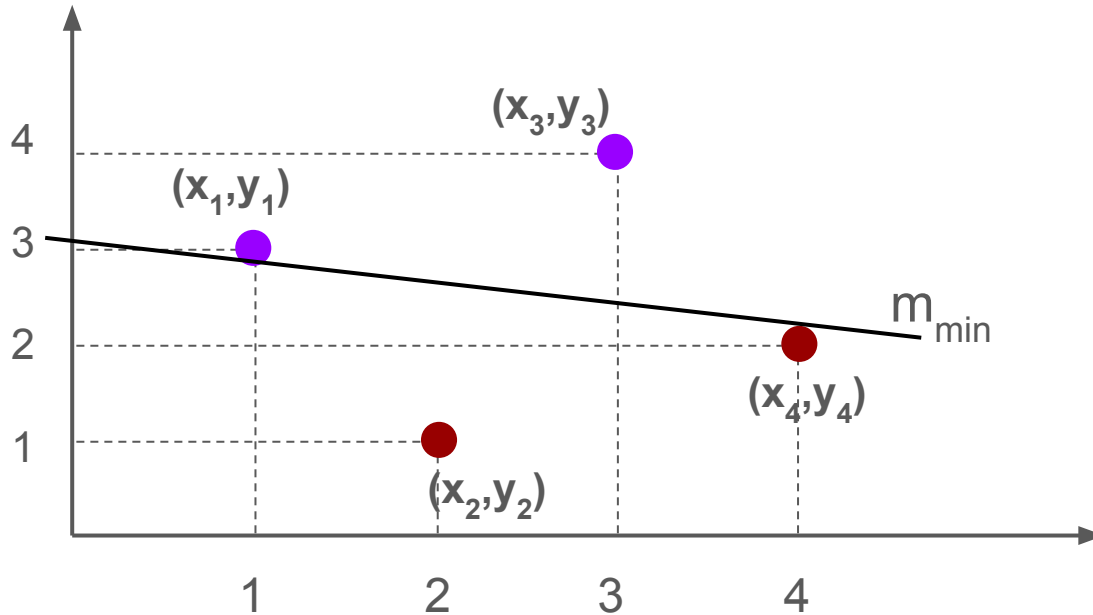
$$m_{\max} = \frac{y_3 - y_2}{x_3 - x_2}$$

$$m_{\max} = \frac{4 - 1}{3 - 2}$$

$$m_{\max} = 3$$

Analytical solution for a linear classifier in $\mathbb{R}^2$

We want to find a line $y = m x + b$ that separates two data classes



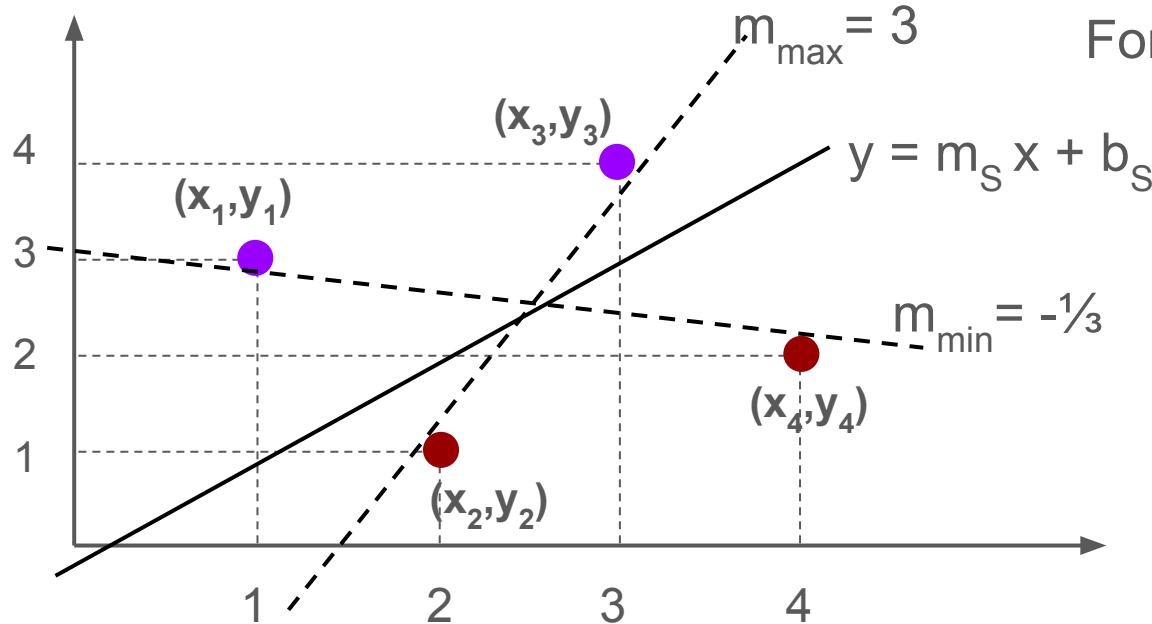
$$m_{\min} = \frac{y_4 - y_1}{x_4 - x_1}$$

$$m_{\min} = \frac{2 - 3}{4 - 1}$$

$$m_{\min} = -1/3$$

Analytical solution for a linear classifier in $\mathbb{R}^2$

We want to find a line $y = m x + b$ that separates two data classes



For any $m_{\max} > m > m_{\min}$

$$\begin{cases} b < 3 - m \\ b > 1 - 2m \\ b < 4 - 3m \\ b > 2 - 4m \end{cases}$$

$$(m_s = 1, 1 > b_s > -1)$$



A learning algorithm for the perceptron (PLA)

- Given a perceptron with weights $\mathbf{w} = [w_0, w_1, w_2, w_3, \dots w_N]^T$ and an activation function $\text{sign}()$, i.e. $\text{sign}(x) = 1$ if $x > 0$, -1 else
- Given inputs $\mathbf{x}_k = [x_0, x_1, x_2, x_3, \dots x_N]^T$ with targets $t_k = \{1, -1\}$
- Then, the output is $y(\mathbf{x}_k) = \text{sign}(\mathbf{w}^T \mathbf{x}_k)$ and the error is $e_k = y(\mathbf{x}_k) - t_k$
- Perceptron learning algorithm (PLA):
 - At step n , randomly select a data sample $\mathbf{x}_k(n)$ and compute $e_k(n)$
 - If $e_k(n) \neq 0$, update weights as $\mathbf{w}(n+1) = \mathbf{w}(n) - e_k(n) \mathbf{x}_k(n)$
 - Repeat until no further classification improvement is achieved



A perceptron learning algorithm exercise

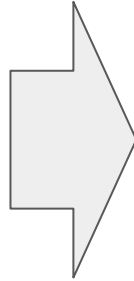
- Given the following four data points and their target binary categories:
 - $P1 = \{(1,3):1\}$, $P2 = \{(2,1):-1\}$, $P3 = \{(3,4):1\}$, $P4 = \{(4,2):-1\}$
- Use a perceptron with three weights $[w_0, w_1, w_2]$ and the perceptron learning algorithm to build a binary classifier
- Consider the initial set of weights $[-3, 0, 2]$ and following sequence of data points $k(n) = 1, 4, 1, 3, 3, 4$ to execute the algorithm

...you should get to a solution after those steps!

A perceptron learning algorithm exercise



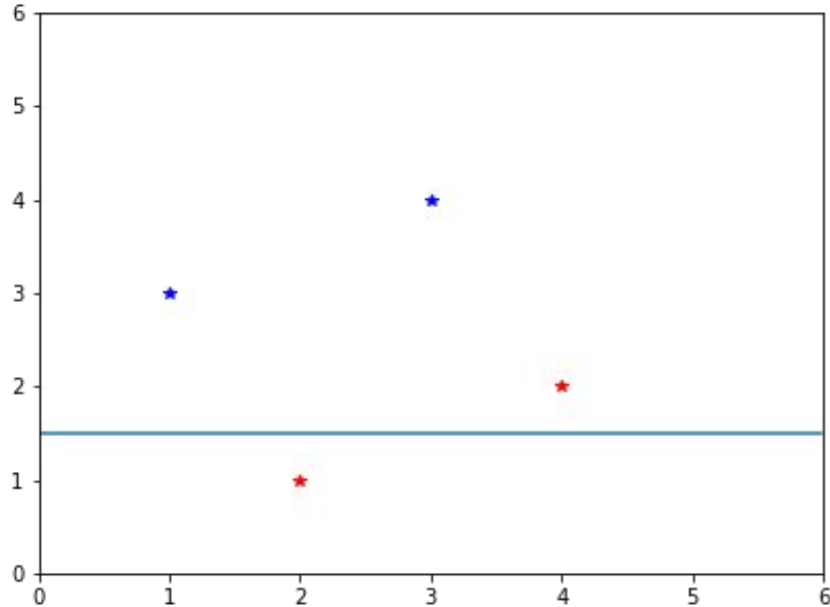
k	x_0	x_1	x_2	t
1	1	1	3	1
2	1	2	1	-1
3	1	3	4	1
4	1	4	2	-1



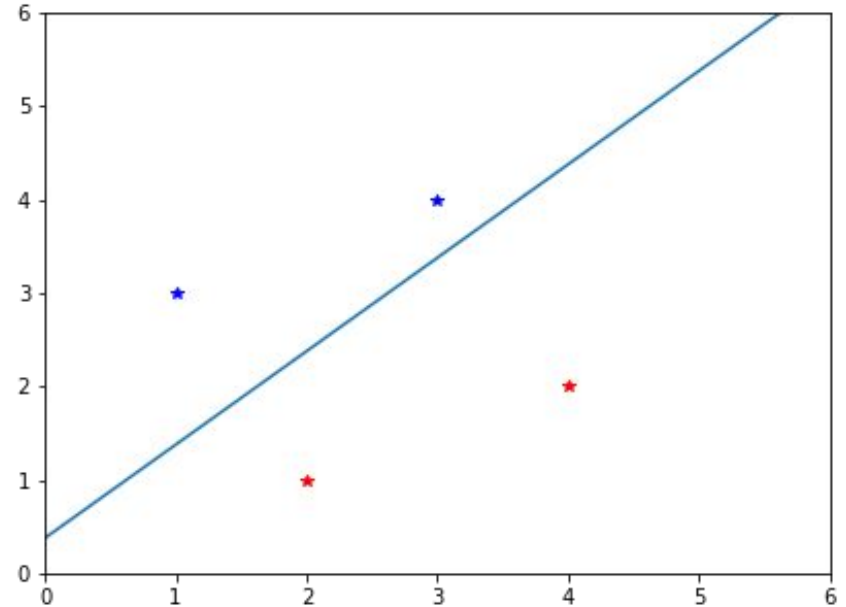
n	k(n)	w_0	w_1	w_2	Σ	y	e
1	1	-3	0	2	3	1	0
2	4	-3	0	2	1	1	2
3	1	-5	-8	-2	-19	-1	-2
4	3	-3	-6	4	-5	-1	-2
5	3	-1	0	12	1	1	0
6	4	-1	0	12	1	1	2
7	-	-3	-8	8	-	-	-



A perceptron learning algorithm exercise



$$n = 1: x_2 = 3/2$$



$$n = 7: x_2 = x_1 + 3/8$$

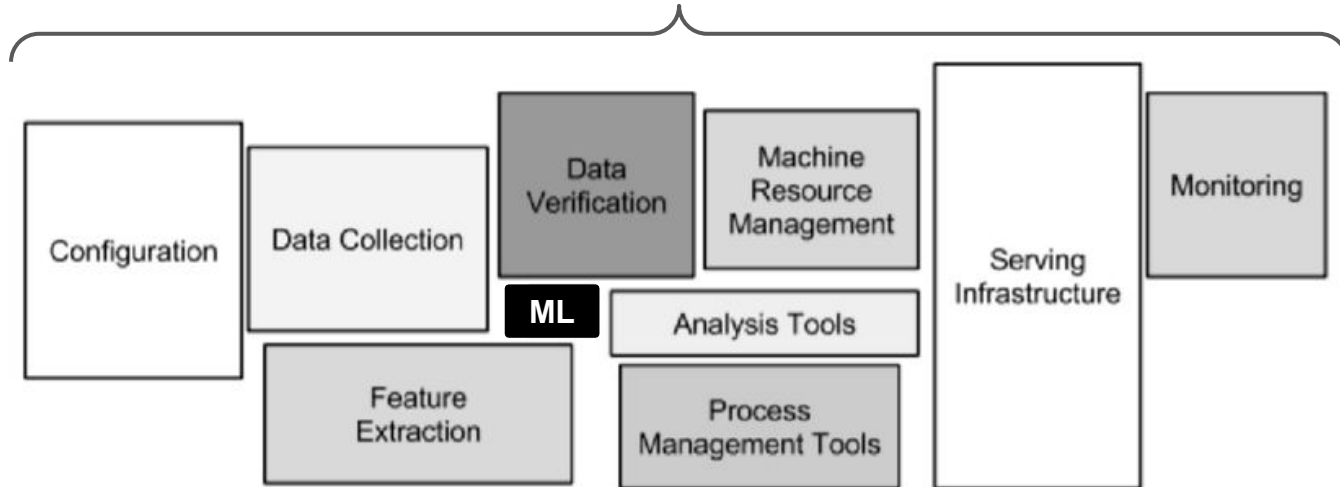
Machine learning in practice

Machine learning vs artificial intelligence:

- Machine learning: system ability to learn a task from a set of observations (learning from data)
- Artificial intelligence: system ability to perform tasks that are normally associated to human intelligence
 - about user's perception of intelligent behaviors
 - does not necessarily requires machine learning

Machine learning in practice

$$\text{AI} = \frac{\beta \text{ UX} + (1-\beta) \text{ ML}}{\text{Software Engineering}} \quad \beta \approx 0.8$$



Machine learning in practice

Hidden technical debt in machine learning systems:*

- Complex models erode boundaries
- Data dependencies cost more than code dependencies
- Feedback loops and ML-system anti-patterns
- Dealing with changes in the external world
- and more...

* <https://proceedings.neurips.cc/paper/2015/file/86df7dcfd896fcac2674f757a2463eba-Paper.pdf>

Machine learning in practice

When to use machine learning?

- The problem at hand is complex in nature
- A closed formula or analytic solution is not available
- A solution cannot be coded by means of rules
- Data is available or, alternatively, can be collected
- The data comprises latent factors to support the solution

Machine learning in practice

Legal and ethical aspects of machine learning:

- Privacy and confidentiality (PII and PHI)
- Bias and discrimination (social, racial, gender, etc.)
- Intellectual property and copyrights (fair use doctrine)
- Regulations (EU AI Act) and standards (ISO/IEC 42001)
- AI Fairness, Responsible AI, Explainable AI, etc...



Hands-on Coding Session W1S1

Hands on coding session W1S1



- Look at the three functions provided in Part 1 and make sure you understand what they do
- Use the provided script to reproduce the perceptron learning algorithm example worked on in class
- Complete the rest of the exercises in Part 1 and Part 2
- Include two term project ideas in Part 3 (homework 1)



Mean Square Error (MSE)

- We need to measure how close $\hat{y}(x)$ gets to the actual $y(x)$
- By minimizing such measure we find the best estimates $\hat{y}(x)$
- We will see more metrics later, but let us start with MSE

$$\text{MSE} = \frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n)^2$$

the actual values y_n

the estimates $\hat{y}_n$

Week 1 - Closing remarks

Wrap-up and Homework 1

- What did we look at today?
- Further reading and support materials
- First homework assignment

What did we look at today?

Syllabus and Course Workplan	• Goals and learning outcomes • Course evaluation plan • Grading policy • Timeline and work plan
Introduction to Machine Learning	• Learning from data • The three learning paradigms • A supervised learning example • Machine learning in practice
Hands-on coding session W1S1	• The perceptron as a binary classifier • Convergence and the learning rate parameter • Two term project ideas... homework!

Further reading and support materials

- James, Witten, Hastie, Tibshirani, Taylor, “An Introduction to Statistical Learning with Applications in Python” (sections 1, 2.1, 2.2) <https://www.statlearning.com/>
- Abu-Mostafa, Magdon-Ismail, Lin, “Learning From Data” (sections 1.1 and 1.2) <https://work.caltech.edu/telecourse>
- Sculley, Holt, Golovin, Davydov, Phillips, Ebner, Chaudhary, Young, Crespo, Dennison, “Hidden Technical Debt in Machine Learning Systems” <https://proceedings.neurips.cc/paper/2015/file/86df7dcfd896fcaf2674f757a2463eba-Paper.pdf>
- El Mestari, Lenzini, Demirci, “Preserving Data Privacy in Machine Learning Systems”, <https://www.sciencedirect.com/science/article/pii/S0167404823005151>
- U.S. Copyright Office Fair Use Index <https://www.copyright.gov/fair-use/index.html>

First homework assignment (due date 06/26)

- Read the following support materials and be ready to answer some questions about them:
 - Hidden Technical Debt in Machine Learning Systems
 - Preserving Data Privacy in Machine Learning Systems
 - U.S. Copyright Office Fair Use Index
- Complete all coding exercises from coding session W1S1
- Write two ideas you want to consider for your Term project. For each, provide: a title, a brief description, and some bullet points about why you want to consider it (include this in your W1S1 notebook, Part 3).



Remember:

Office Hours

Tuesday 06/24 5:30-6:30pm

Next Class

Thursday 06/26, 5:30-9:30pm

First Homework Due

Thursday 06/26, 11:59pm